MATH19622-1E2: Introduction to Probability and Statistics

4 Lecture Series (90 minutes per Lecture)

Course Overview and Descriptive Statistics

Anthony R. Thornton

2025/2026

Outline - Next Section I

History of the course

This course was newly developed by **Anthony R. Thornton** for the 2025/2026 academic year.
It is based on:

Mathematics 2M1 — Probability and Statistics written by Robert Gaunt, School of Natural Sciences, University of Manchester, and David J. Silvester, School of Natural Sciences, University of Manchester.

Special thanks to Google Gemini for proofreading and structural suggestions.

Optional Books I

Optional **Open Access** additional reading for this course:

Primary Resources

- **Module 1 (Descriptive Statistics):**
  Introductory Statistics by **OpenStax**.
  `https://openstax.org/details/books/introductory-statistics`
- **Modules 2–4 (Probability):**
  Introduction to Probability, Statistics, and Random Processes by **H. Pishro-Nik**. (Tailored for EEE).
  `https://www.probabilitycourse.com`

Optional Books II

Supplementary / General Reference

- *OpenIntro Statistics* by **Diez, Çetinkaya-Rundel, & Barr**.
  (A clean, concise alternative for general statistics concepts).
  `https://www.openintro.org/book/os/`
- *Think Stats* by **Allen B. Downey**.
  (Teaches statistics through Python programming).
  `https://greenteapress.com/wp/think-stats-2e/`
- **Khan Academy (Statistics & Probability)**.
  (Video tutorials useful for visualising histograms and distributions).
  `https://www.khanacademy.org/math/statistics-probability`

Optional Books III

For the Mathematically Interested

Introduction to Probability by **Grinstead & Snell**.
(Classic text with rigorous proofs and derivations).
`https://math.dartmouth.edu/~prob/prob/prob.pdf`

Course Structure

This course is designed to cover the final part of Mathematics for EEE 1E2, namely:

- **Statistics** Discrete and continuous data. Mean, mode, and median. Range, variance and **standard deviation**. Frequency curves: Normal distribution and standardised normal.
- **& Probability** Empirical and classical. Addition and multiplication laws. Probability distributions. Mean and standard deviation. Binomial and Poisson distribution. Normal distribution curves.

It will consist of four approximately 90-minute lectures.

The first two elements: Differential Vector Calculus and Integral Vector Calculus are **covered** by a separate course taught by Igor Chernyavsky.

Assessment

The assessment for these elements of the course consists of:

- **Online Coursework** (Week 10)
  Contributes **10%** to the final grade.

- **End-of-Semester Examination**
  Contributes **80%** to the final grade.

The remaining **10%** is assessed in the section taught by **Igor Chernyavsky**.

Note both these test wills also include questions from the earlier parts 1 and 2.

Course Overview

Overview: 5 Thematic Blocks delivered across 4 Lectures (100 mins each).

Session	Thematic Blocks	Learning Outcomes
Lecture 1	Overview of Part **Block 1:** Descriptive Statistics	 ILO 5: Data Analysis
Lecture 2	**Block 2:** Prob. Fundamentals **Block 3:** Discrete Distributions	ILO 6: Modeling
Lecture 3	**Block 4:** Continuous Distributions	ILO 6: Problem Solving
Lecture 4	**Block 5:** Linear Regression	ILO 5: Interpretation

Accessing the Living Course Materials

While official PDFs are on **Canvas**, the ‘Living’ version of this course (including live Python code and LaTeX source) is hosted on GitHub.

GitHub Repository

https://github.com/antRthorn/UoM-MATH19622-Public

How to get the code (for the interested):

1. **Clone the Repo:**

```
git clone https://github.com/antRthorn/UoM-MATH19622-Public.git
```

2. **Run the Examples:**

```
uv run path/to/script.py
```

Note: Use the VS Code Dev Container for a zero-install Python environment.

The QA Bounty Program

- **The Task:** Find a typo, a calculation error, or suggest a major improvement to the course code or examples.
- **The Rewards:**
  - **£5 Voucher:** For minor errors (typos, broken links).
  - **£10 Voucher:** For significant clarity improvements or bug fixes.
- **The Process:** Submit a **Pull Request** or an **Issue** on GitHub.
- **The Catch:** First person to report the specific error wins the bounty for that slide!

'An engineer who never makes a mistake has never made anything.'

Block I

Descriptive Statistics

Outline - Next Section I

Outline - Next Section II

Why Statistics for EEE?

As Electrical Engineers, you deal with **signals** and **systems**.

- **The Ideal World:** $V = IR$. A perfect resistor has exactly $100\,\Omega$ resistance.
- **The Real World:** A resistor has a tolerance (e.g., $100 \pm 5\,\Omega$). Thermal noise introduces random fluctuations in voltage.

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The Goal

Statistics allows us to quantify this uncertainty, distinguish the **signal** from the **noise**, and build reliable systems from imperfect components.

Outline - Next Section I

Outline - Next Section II

Key Definitions: Population vs. Sample

Definition (Population)

The entire set of items/individuals of interest.

- Example: *Every* capacitor produced by a factory in 2025.
- Measures are called **Parameters** (e.g., population mean, μ).

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A subset of the population selected for analysis.

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In this course, we mostly calculate Statistics to estimate Parameters.

Representative Sample I

To draw valid conclusions about a population, our sample must be fair.

Definition (Representative Sample)

A representative sample is a subset that accurately reflects the characteristics of the larger **population**.

The Goal: Randomness To be representative, samples are usually collected randomly.

Definition (Simple Random Sample)

Every member of the population has an **equal chance** of being selected.

To achieve this we have to remove human choice and systematic errors.

Representative Sample II

The Pitfall: Bias

Bias occurs when the sample systematically favours certain outcomes.

- *Example:* Testing only the first 100 chips produced in the morning.
- *Issue:* This ignores defects caused by machine overheating later in the day.
- *Consequence:* Your failure rate is underestimated.

Discussion: Is this Representative? I

Example (The Solar Engineering)

Scenario: An engineering team is designing a new solar power inverter. To test its reliability, they install 50 prototype units on rooftops in **Manchester, UK** and monitor them for a year.
The data shows a 99.8% reliability rate. The team concludes the inverter is ready for a global rollout, including markets in **Dubai** and **Arizona**.
Question: Is this sample representative of the global operating environment?

- Why or why not?
- What variables might be missing from the sample?

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Discussion: Is this Representative? II

Solution (The Problem: Environmental Bias)

No. *The sample is biased towards a cool, cloudy climate.*

- ***Missing Variable:*** *Extreme heat ($> 40^\circ C$) effectively derates electronics and stresses cooling systems.*
- ***Consequence:*** *High failure rates possible in hot climates.*

The Problem of ‘Luck’ (Sampling Error) I

Even if our sample is perfectly random and representative (no bias), it is still a subset. By pure chance, we might pick slightly more ‘high’ values than ‘low’ values.

Definition (Sampling Variation (Random Error))

Random error is the difference between the sample statistic ($\bar{x}$) and the true **population parameter** (μ) caused purely by chance.

Example

You flip a fair coin 10 times. You *should* get 5 Heads.

- **Result:** You get 8 Heads.
- **Is the coin biased?**

The Problem of 'Luck' (Sampling Error) II

The Solution: Confidence Intervals

Because of this variation, we rarely state a single number. Instead, we calculate a range:

'The voltage is $5.0\text{V} \pm 0.1\text{V}$ with 95% confidence.'

Key Property: Sample Size

Unlike Bias, random error **decreases** as the sample size (n) increases. *(Law of Large Numbers)*

Ethics in Statistics: The 'P-Hacking' Problem I

Since 'bad luck' (random variation) happens, can we exploit it?

Example (The (Un)ethical Engineer)

An engineer wants to prove a new battery lasts longer.

- They run **100 separate small experiments**.
- In 95 cases, the new battery is no better than the old one.
- In 5 cases, by pure random chance, the new battery performs amazingly.
- **Action:** They publish *only* the 5 'successful' results and hide the rest.

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Ethics in Statistics: The ‘P-Hacking’ Problem II

This is P-Hacking (Data Dredging)

Technically, the data isn’t ‘fake’; the measurements really happened. However, omitting the failures distorts the probability reality.

Rule: You must report **all** attempts, not just the ones that fit your desired conclusion.

Outline - Next Section I

Outline - Next Section II

Where does Data come from?

Definition (Data)

Data is the set of observations or measurements we collect.

In Engineering, it comes from two main sources:

1. Finite Population

A fixed group of existing items.

- **Example:** A batch of 500 resistors in a box.
- **Goal:** Characterise this specific batch.

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A system generating data over time.

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*In both cases, we use **samples** (small datasets) to estimate the properties of the whole.*

Classifying Data

Data generally falls into two broad categories:

1. **Qualitative (Categorical):** Non-numerical descriptions.
   - Examples: Component colour codes, Pass/Fail status.

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1. **Qualitative (Categorical):** Non-numerical descriptions.
   - Examples: Component colour codes, Pass/Fail status.
2. **Quantitative (Numerical):** Measured or counted values.
   - This is our main focus in EEE.
   - Further split into **Discrete** and **Continuous**.

Discrete vs. Continuous Data

1. Discrete Data

- Countable values (integers).
- No intermediate values exist.
- **EEE Examples:**
  - Number of packets lost in transmission.
  - Number of electrons (at quantum scale).
  - Digital logic levels (0 or 1).

2. Continuous Data

- Measurable values (real numbers).
- Can take any value within a range.
- **EEE Examples:**
  - Voltage across a capacitor.
  - Current in a wire.
  - Signal propagation delay.

Outline - Next Section I

Outline - Next Section II

Measures of Central Tendency

We often want to summarise a dataset with a single value that represents the **centre** or **location** of the data.

The three most common measures are:

1. **Mean:** The arithmetic average.
2. **Median:** The middle value.
3. **Mode:** The most frequent value.

1. The Arithmetic Mean

The mean is the most common measure of centre. It is the sum of all values divided by the number of observations.

Definition (Sample Mean ($\bar{x}$))

For a sample of size n with values $x_1, x_2, \ldots, x_n$:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i \tag{1}$$

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Recipe (Calculating the Mean)

1. Sum all the values.
2. Divide by the number of values.

Note:

- For a **Population**, we use the symbol μ.
- The mean is highly sensitive to **outliers** (extreme values).

2. The Median

Definition (Median)

The median is the middle value when the data is ordered from smallest to largest.

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Recipe (Calculating the Median)

1. Sort the data: $x_{(1)} \leq x_{(2)} \leq \cdots \leq x_{(n)}$.
2. Find the position:
   - If n is **odd**: The median is the exact middle number.
   - If n is **even**: The median is the mean of the two middle numbers.

*The median is **robust** (less affected by outliers).*

3. The Mode

Definition (Mode)

The mode is the value that occurs most frequently in the dataset.

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Recipe (Calculating the Mode)

1. List all unique values in the dataset.
2. Count the frequency (number of occurrences) for each value.
3. Identify the value(s) with the highest frequency.
    - If multiple values share the highest count, the data is **multimodal**.
    - If all values appear only once, there is **no mode**.

- Useful for Qualitative data (e.g., 'The most common failure mode is overheating').

Worked Example: Signal Latency I

Example (Network Router Latency)

An engineer measures the latency (in ms) of 5 packets passing through a network router:

Data: 12, 15, 12, 200, 14

Questions:

1. Calculate the mean, median, and mode.
2. Draw an engineering conclusion: Which measure best represents the network's typical performance?

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Worked Example: Signal Latency II

Solution

Part 1: Calculations

- ***Mean:***

$$\bar{x} = \frac{12 + 15 + 12 + 200 + 14}{5} = \frac{253}{5} = \boldsymbol{50.6\ ms}$$

- ***Median:*** *Sort the data:* $12, 12, \mathbf{14}, 15, 200.$

$$Median = \boldsymbol{14\ ms}$$

- ***Mode:*** *The most frequent value is* ***12 ms****.*

Worked Example: Signal Latency III

Solution

Part 2: Interpretation

- ***Mean (50.6 ms):*** *Heavily skewed by the single outlier (200 ms). This metric falsely suggests the entire network is slow, which is only true for 1 packet (perhaps due to a temporary buffer spike).*
- ***Median (14 ms):*** *Gives a much more accurate representation of the 'typical' performance the user will experience.*

Key Takeaway

Always look at the data distribution. If the data is skewed (like network latency, income, or component lifespans often are), the **median** is a much safer measure of the centre than the **mean**.

Interactive Demonstration

The Pull of the Outlier

Let's look at how the Mean and Median react in real-time when we introduce an extreme network delay.

Launch Python Simulation

(Note: This will open an external interactive Matplotlib window.)

Outline - Next Section I

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Our Python Environment: Keeping it Standard

The Engineering Stack

- **Editor:** Visual Studio Code (VSCode)
- **Environment:** Devcontainers (Isolated and reproducible)
- **Package Manager:** `uv` (Blazing fast, modern standard)

Getting Ready for Statistics: For this course we need *numpy*, *scipy* and *matplotlib*.

Setup

```
# In your VSCode terminal inside the devcontainer:
uv add numpy scipy matplotlib
```

No PyCharm, no Anaconda, no conflicts. Just engineering efficiency!

Statistics in Action: Python I

In modern engineering, we rarely calculate statistics by hand. We leverage the power of the **SciPy Stack**, specifically `numpy` and `scipy.stats`.

NumPy (The Foundation)

- **Efficiency:** Highly optimised for N-dimensional arrays.
- **Speed:** Vectorised operations outperform standard Python loops.
- **Linear Algebra:** Essential for handling large datasets.

SciPy (The Toolbox)

- **scipy.stats:** Specialised functions for probability distributions (Normal, t-tests, p-values).
- **Fitting:** Tools to fit experimental data to mathematical models.
- **Integration:** Built-in solvers for complex engineering calculus.

Statistics in Action: Python II

Python Code Example: Mean, mode, medium

```
import numpy as np
from scipy import stats
# Our dataset (e.g., latency in ms)
data = [12, 15, 12, 200, 14]
# Calculate Measures
mean_val = np.mean(data) # Output: 50.6
median_val = np.median(data) # Output: 14.0
mode_val   = stats.mode(data, keepdims=True)
print(f"Mean: {mean_val}")
print(f"Median: {median_val}")
# Access the first element [0] to get the value, not the
    count
print(f"Mode: {mode_val.mode[0]}")
```

Statistics in Action: Python III

- **Imports:** We use `numpy` (standard for arrays) and `scipy.stats` (for specific statistical functions).
- **Efficiency:** `np.mean()` and `np.median()` are highly optimised. They run much faster than writing a loop yourself.
- **The 'Mode' Trap:** Unlike mean/median, `stats.mode` returns a complex object (Value + Count).
- **Extraction:** We must use `.mode[0]` to pull out the actual number for printing.

Outline - Next Section I

Outline - Next Section II

Summary - First Half

- **Data Types:** Identify if your variables are Discrete (counts) or Continuous (measurements).
- **Notation:** $\bar{x}$ for Sample Mean, μ for Population Mean.
- **Central Tendency:**
  - **Mean:** Mathematical centre, sensitive to outliers.
  - **Median:** Positional centre, robust.
  - **Mode:** Most frequent, good for categories.

Next: We will look at *Measures of Dispersion* (Range, Variance, Standard Deviation) to understand the “spread” of the data.

Outline - Next Section I

Outline - Next Section II

Why Central Tendency isn't enough

Example (The 'Average' Weather)

Consider two cities with the same Mean Yearly Temperature (20°C).

Question: Would you build the same bridge or roads in both?

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- **City A (Stable):** Ranges from 18°C to 22°C.
- **City B (Extreme):** Ranges from -10°C to 50°C.

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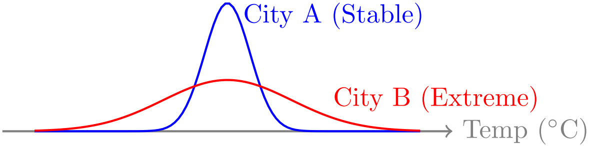
An engineer needs to design for the **Extremes** (Spread).



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Why do we measure Spread?

The Danger of the 'Average'

In engineering design, the average is often a completely useless metric for safety. We must design for the extremes.

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Real-World Engineering Constraints

- A structural beam fails at the **maximum** load, not the average load.
- A circuit overheats at the **peak** current, not the average current.

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The Meaning of Dispersion

Dispersion tells us how reliable or volatile our system is.

$\Longrightarrow$ **High Dispersion = High Risk or Low Precision.**

1. The Range

The simplest measure of spread.

$$\text{Range} = x_{max} - x_{min}$$

Weakness: Sensitivity to Outliers

Dataset: $\{1, 2, 3, 4, 5\} \rightarrow \text{Range} = 4$.
Dataset: $\{1, 2, 3, 4, \mathbf{100}\} \rightarrow \text{Range} = 99$.

The range ignores the internal structure of the data. It only cares about the two most extreme values, which might be errors.

2. Interquartile Range (IQR)

If the Median is our robust 'average', the IQR is our robust 'spread'. It measures the range of the **middle 50%** of your data, completely ignoring the extremes.

Definition (Quartiles)

To find the IQR, we split the sorted data into four equal quarters:

- Q_1 **(Lower Quartile):** The 25th percentile.
- Q_2 **(Median):** The 50th percentile.
- Q_3 **(Upper Quartile):** The 75th percentile.

The IQR Formula

$$\text{IQR} = Q_3 - Q_1$$

By stripping away the bottom 25% and top 25% of the data, the IQR becomes highly immune to sensor glitches and extreme outliers.

2. The Variance (σ^2 or s^2)

How do we measure 'average spread' using *all* data points? **The Logic:**

1. Find the distance of every point from the Mean: $(x_i - \bar{x})$.
2. Square them to remove negatives: $(x_i - \bar{x})^2$.
3. Average these squared distances.

Why Square? If we just added the distances $(x - \bar{x})$, the negatives would cancel the positives, summing to zero. Squaring makes everything positive.

Population vs. Sample Variance

This is the most common mistake in engineering statistics.

Population Variance (σ^2)

Used when you have **all** the data (e.g., a census).

$$\sigma^2 = \frac{\sum(x-\mu)^2}{N}$$

Sample Variance (s^2)

Used when you have a **subset** (e.g., 10 sensors out of 1000).

$$s^2 = \frac{\sum(x-\bar{x})^2}{n-1}$$

Why $n-1$? This is **Bessel's Correction**. It corrects for the fact that samples tend to underestimate the true spread of the population.

3. Standard Deviation (σ or s)

The Problem with Variance: If we measure length in meters (m), Variance is in meters-squared (m^2). This is physically confusing.
The Solution: Take the square root to return to the original units.

$$s = \sqrt{s^2} = \sqrt{\frac{\sum(x - \bar{x})^2}{n - 1}}$$

- **High SD:** Data is spread out (Noisy, Unreliable).
- **Low SD:** Data is clustered near the mean (Precise, Consistent).

Recipe (Calculating Variance (s^2) & Standard Deviation (s))

1. Calculate the **Mean** ($\bar{x}$) of the dataset.
2. Find the **deviation** for each data point by subtracting the mean: $(x_i - \bar{x})$.
3. **Square** each deviation to make it positive: $(x_i - \bar{x})^2$.
4. **Sum** all the squared deviations to get the 'Sum of Squares' (SS).
5. Divide the sum by $n - 1$ to find the **Sample Variance** (s^2).
   - *Note: Dividing by $n - 1$ (Bessel's Correction) is standard for engineering samples.*
6. Take the **square root** of the variance to find the **Standard Deviation** (s).

Worked Example (Hand Calc) I

Example (Calculating standard deviation and variance)

Compute the standard deviation and variance of the following data.
Data: $\{2, 4, 4, 4, 5, 5, 7, 9\}$. $(n = 8)$.

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Worked Example (Hand Calc) II

Solution

Step 1: Calculate Mean $\bar{x} = 40/8 = 5$.

x	*Step 2: Deviation* $(x - \bar{x})$	*Step 3: Squared* $(x - \bar{x})^2$
2	*-3*	*9*
4	*-1*	*1*
...	...	...
9	*+4*	*16*
Sum *(Step 4)*	***0***	***30***

Step 5: $s^2 = \frac{30}{8-1} = \frac{30}{7} = 4.29$

Step 6: $s = \sqrt{4.29} \approx 2.07$

Python Implementation

Warning: NumPy calculates **Population** variance by default (divides by n). You must tell it to use the $n-1$ correction.

Calculating SD in Python

```
import numpy as np

data = [2, 4, 4, 4, 5, 5, 7, 9]

# 1. Population SD (divides by n) - RARELY USED
pop_sd = np.std(data)

# 2. Sample SD (divides by n-1) - ENGINEERING STANDARD
# ddof = Delta Degrees of Freedom
sample_sd = np.std(data, ddof=1)

print(f"Sample Std Dev: {sample_sd:.2f}")
```

Variance vs. Standard Deviation

Variance (σ^2)

The Mathematical Tool

- **Units:** Squared (e.g., m^2, s^2, V^2).
- **Pro:** Mathematically clean. Variances can be added together directly.
- **Con:** Hard to interpret physically.

$$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$$

Standard Deviation (σ)

The Reporting Tool

- **Units:** Original (e.g., m, s, V).
- **Pro:** Intuitive. Describes the 'average distance' from the mean.
- **Con:** The math is messy (Square roots don't add).

$$\sigma_{total} = \sqrt{\sigma_1^2 + \sigma_2^2}$$

Key Takeaway: Calculate with Variance, communicate with Standard Deviation.

Outline - Next Section I

Outline - Next Section II

Lecture Summary: Descriptive Statistics I

We use statistics to distinguish the **Signal** (Location) from the **Noise** (Dispersion).

1. Location (The Signal)

- **Mean ($\bar{x}$):** The arithmetic centre. *Warning: Sensitive to outliers.*
- **Median:** The middle value. *Use for skewed data (e.g., Latency).*
- **Mode:** The most frequent value.

2. Dispersion (The Noise)

- **Variance (s^2) & Std Dev (s):** The math & reporting tools.
- **IQR:** The middle 50%. Highly robust.
- **Range:** Just $x_{max} - x_{min}$. Fragile.

Lecture Summary: Descriptive Statistics II

The Golden Rule: Bessel's Correction

When working with a **Sample** (which is almost always), you must divide by $n-1$ to avoid underestimating the spread.

$$s = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n-1}}$$

Next Lecture: We move from describing data to predicting it (Probability).

MATH19622-1E2: Introduction to Probability and Statistics

4 Lecture Series (90 minutes per Lecture)

Probability Fundamentals and Discrete Models

Anthony R. Thornton

2025/2026

Block II

Probability Fundamentals

Recap: Descriptive Statistics

Last week, we learned how to summarise historical data by separating the **Signal** from the **Noise**.

1. Location (The Signal)

Where is the center of the data?

- **Mean:** The mathematical center.
- **Median:** The robust middle (ignores outliers).

2. Dispersion (The Noise)

How spread out is the data?

- **Variance/Std Dev:** The mathematical spread (remember $n - 1$ for samples!).
- **IQR:** The robust middle 50%.

The Transition: Descriptive statistics let us understand the **past**. Today, we begin using Probability to predict the **future**.

Outline - Next Section I

Outline - Next Section I

Outline

- **Definitions & Notation**
  - Sample Space (S), Events, Complement ($P(A') = 1 - P(A)$).
- **The Addition Law (Unions)**
  - Mutually Exclusive: $P(A \cup B) = P(A) + P(B)$.
  - General Rule: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- **The Multiplication Law (Intersections)**
  - Independent Events: $P(A \cap B) = P(A) \times P(B)$.
  - Conditional Probability: $P(A \cap B) = P(A) \times P(B|A)$.

Outline - Next Section I

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The Sample Space and Events I

Definition (Experiment and Sample Space)

An **experiment** is any process whose outcome is subject to uncertainty.
The **sample space** (S) is the set of all possible outcomes.

Example (Testing Microcontrollers)

Suppose we test 3 microcontrollers off a production line. Each either Passes (P) or Fails (F). The sample space has 8 possible outcomes:

$$S = \{PPP, PPF, PFP, PFF, FPP, FPF, FFP, FFF\}$$

Definition (Event)

An **event** is any subset of the sample space.

The Sample Space and Events II

If we define Event A as 'exactly one microcontroller fails', then:

$$A = \{PPF, PFP, FPP\}$$

Set Operations in Probability

Events are usually combined using standard set operations. Let A and B be events in sample space S.

- **Complement ($\bar{A}$ or A'):** All outcomes *not* in A.
  - $P(\bar{A}) = 1 - P(A)$
- **Intersection ($A \cap B$):** Outcomes in both A **AND** B.
- **Union ($A \cup B$):** Outcomes in A **OR** B (or both).
- **Mutually Exclusive ($A \cap B = \varnothing$):** Events that cannot happen at the same time. The intersection is the empty set ($\varnothing$).

The Addition Law (Unions)

How do we calculate the probability of A **OR** B happening?

1. For Mutually Exclusive Events

If A and B cannot happen together ($A \cap B = \varnothing$), we simply add their probabilities:

$$P(A \cup B) = P(A) + P(B)$$

The Addition Law (Unions)

How do we calculate the probability of A **OR** B happening?

1. For Mutually Exclusive Events

If A and B cannot happen together ($A \cap B = \varnothing$), we simply add their probabilities:

$$P(A \cup B) = P(A) + P(B)$$

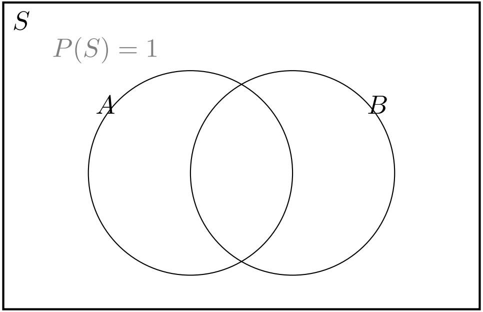
2. The General Addition Rule

If events *can* happen together, simply adding them will count the intersection twice. We must subtract the overlap:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Visualising the Addition Law

The addition law: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
Venn diagrams make the subtraction intuitive.



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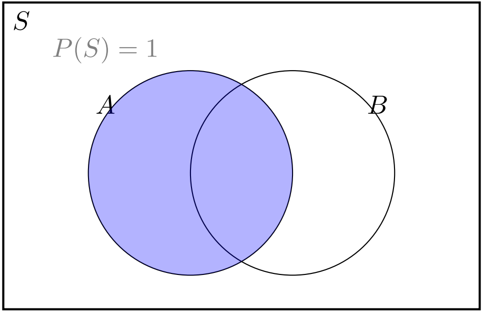

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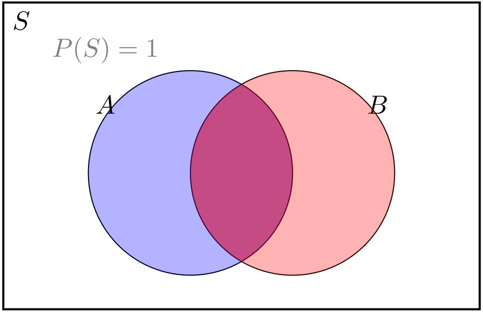

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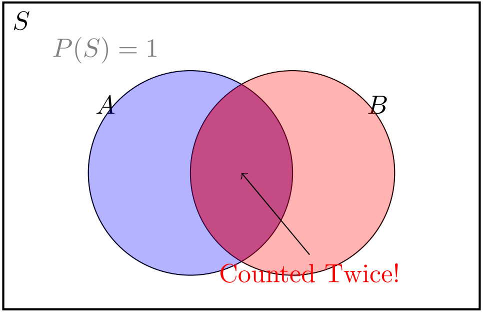

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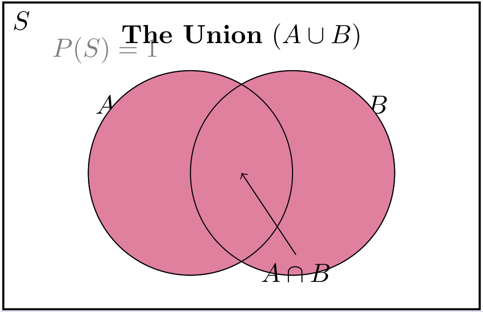

Why do we subtract?

Adding $P(A) + P(B)$ counts the intersection **twice**. To fix this, we must subtract one instance of the intersection:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

A Simple Dice Example

Example (The 6-Sided Die)

Consider rolling a fair 6-sided die. Let event A be rolling an even number, and let event B be rolling a number divisible by 3.

Questions:

1. Write down the elements of subsets A and B.
2. Calculate $P(A)$ and $P(B)$.
3. Write down the elements of the intersection (the event that the number is both even **and** divisible by 3) and calculate its probability, $P(A \cap B)$.
4. Calculate the probability that the number rolled is either even **or** divisible by 3, $P(A \cup B)$, by counting the elements.
5. Confirm the General Addition Law for this case.
6. Are events A and B mutually exclusive?

Solution: A Simple Dice Example (1/3)

Solution

1. Elements of A and B:

- *Sample space* $S = \{1, 2, 3, 4, 5, 6\}$
- A *(Even numbers)* $= \{2, 4, 6\}$
- B *(Divisible by 3)* $= \{3, 6\}$

Solution: A Simple Dice Example (1/3)

Solution

1. Elements of A and B:

- *Sample space* $S = \{1, 2, 3, 4, 5, 6\}$
- A *(Even numbers)* $= \{2, 4, 6\}$
- B *(Divisible by 3)* $= \{3, 6\}$

2. Calculate $P(A)$ and $P(B)$:

- $P(A) = \frac{3}{6} = \frac{1}{2}$
- $P(B) = \frac{2}{6} = \frac{1}{3}$

Solution: A Simple Dice Example (2/3)

Solution

3. Intersection $P(A \cap B)$***:***

- *The only number both even* ***and*** *divisible by 3 is 6.*
- $A \cap B = \{6\}$
- $P(A \cap B) = \frac{1}{6}$

Solution: A Simple Dice Example (2/3)

Solution

3. Intersection $P(A \cap B)$***:***

- *The only number both even* ***and*** *divisible by 3 is 6.*
- $A \cap B = \{6\}$
- $P(A \cap B) = \frac{1}{6}$

4. Union $P(A \cup B)$ ***by counting:***

- *Elements in A* ***or*** *B (or both) are* $\{2, 3, 4, 6\}$.
- *There are 4 elements, so* $P(A \cup B) = \frac{4}{6} = \frac{2}{3}$.

Solution: A Simple Dice Example (3/3)

Solution

5. Confirming the Addition Law:

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= \frac{3}{6} + \frac{2}{6} - \frac{1}{6} \\ &= \frac{4}{6} = \frac{2}{3} \quad \textbf{\textit{(Matches!)}} \end{aligned}$$

Solution: A Simple Dice Example (3/3)

Solution

5. Confirming the Addition Law:

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= \frac{3}{6} + \frac{2}{6} - \frac{1}{6} \\ &= \frac{4}{6} = \frac{2}{3} \quad \textbf{\textit{(Matches!)}} \end{aligned}$$

6. Are they mutually exclusive?

- ***No.*** *Mutually exclusive events cannot happen at the same time ($A \cap B = \varnothing$). Because rolling a 6 satisfies both conditions ($A \cap B = \{6\}$), they are not mutually exclusive.*

Conditional Probability

Often, knowing that one event has happened changes the probability of another event happening.

Definition (Conditional Probability)

The probability of A given that B has already occurred is:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \text{for } P(B) > 0$$

Independence and The Multiplication Law

Definition (Independent Events)

Two events are **independent** if knowing one has occurred doesn't change the probability of the other.

$$P(A|B) = P(A) \quad \text{and} \quad P(B|A) = P(B)$$

The Multiplication Law (Intersections)

To find the probability of A **AND** B happening:

- **General Rule:** $P(A \cap B) = P(A) \times P(B|A)$
- **If Independent:** $P(A \cap B) = P(A) \times P(B)$

Example of Non-Independence

Example (Drawing Socks without Replacement)

You have 4 red socks and 2 black socks in a basket. You draw two socks from the basket **without replacement**.
Let A be the event of drawing a red sock on your first attempt.
Let B be the event of drawing a red sock on your second attempt.

Questions:

1. Calculate $P(A)$.
2. Assume event A occurred. Calculate the probability of drawing a second red sock, $P(B|A)$.
3. Calculate the probability of getting a red pair, $P(A \cap B)$.
4. Calculate the unconditional probability of drawing a red sock on the second attempt, $P(B)$. Is it the same as $P(B|A)$? *(Hint: What if the first sock was black?)*

Solution: Non-Independence (1/2)

Solution

1. First Draw, $P(A)$***:***

- *Total socks* $= 6$. *Red socks* $= 4$.
- $P(A) = \frac{4}{6} = \frac{2}{3}$

Solution: Non-Independence (1/2)

Solution

1. First Draw, $P(A)$***:***

- *Total socks = 6. Red socks = 4.*
- $P(A) = \frac{4}{6} = \frac{2}{3}$

2. Second Draw (Given A ***occurred),*** $P(B|A)$***:***

- *Because we did not replace the first sock, the physical sample space has changed!*
- *Total socks left = 5. Red socks left = 3.*
- $P(B|A) = \frac{3}{5}$

Solution: Non-Independence (2/2)

Solution

3. The Red Pair, $P(A \cap B)$***:***

To find the probability of both events happening together, we use the ***General Multiplication Law****.*

Because the events are dependent, we multiply the probability of the first event by the conditional probability of the second:

$$P(A \cap B) = P(A) \times P(B|A)$$

Solution: Non-Independence (2/2)

Solution

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To find the probability of both events happening together, we use the ***General Multiplication Law****.*

Because the events are dependent, we multiply the probability of the first event by the conditional probability of the second:

$$P(A \cap B) = P(A) \times P(B|A)$$

- $P(A \cap B) = \frac{2}{3} \times \frac{3}{5} = \frac{6}{15}$
- *Simplifying the fraction gives our final answer:*

$$\boldsymbol{\frac{2}{5}} \textbf{\textit{ (or 40\%)}}$$

Solution: Non-Independence (3/4)

Solution

4. The Trap: Calculating Unconditional $P(B)$

To find the total probability of the second sock being red, we must account for ***both*** *possible pasts.*

- ***Path 1:*** *First was Red* ***AND*** *Second is Red* $(A \cap B)$.
- ***Path 2:*** *First was Black* ***AND*** *Second is Red* $(A' \cap B)$.

Solution: Non-Independence (3/4)

Solution

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- ***Path 1:*** *First was Red* ***AND*** *Second is Red* $(A \cap B)$.
- ***Path 2:*** *First was Black* ***AND*** *Second is Red* $(A' \cap B)$.

We already found Path 1: $P(A \cap B) = \frac{4}{6} \times \frac{3}{5} = \frac{12}{30}$.

Solution: Non-Independence (3/4)

Solution

4. The Trap: Calculating Unconditional $P(B)$

To find the total probability of the second sock being red, we must account for ***both*** *possible pasts.*

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We already found Path 1: $P(A \cap B) = \frac{4}{6} \times \frac{3}{5} = \frac{12}{30}$.
Now for Path 2: $P(A' \cap B) = \frac{2}{6} \times \frac{4}{5} = \frac{8}{30}$.

Solution: Non-Independence (3/4)

Solution

4. The Trap: Calculating Unconditional $P(B)$

To find the total probability of the second sock being red, we must account for ***both*** *possible pasts.*

- ***Path 1:*** *First was Red* ***AND*** *Second is Red* $(A \cap B)$.
- ***Path 2:*** *First was Black* ***AND*** *Second is Red* $(A' \cap B)$.

We already found Path 1: $P(A \cap B) = \frac{4}{6} \times \frac{3}{5} = \frac{12}{30}$.
Now for Path 2: $P(A' \cap B) = \frac{2}{6} \times \frac{4}{5} = \frac{8}{30}$.

Add the two mutually exclusive paths together (Law of Total Probability):

$$P(B) = \frac{12}{30} + \frac{8}{30} = \frac{20}{30} = \mathbf{\frac{2}{3}}$$

Solution: Non-Independence (4/4)

Solution

Comparing $P(B)$ ***and*** $P(B|A)$***:***

Let's look at the two numbers we just calculated:

- *Unconditional probability of drawing a red sock second:* $P(B) = \frac{2}{3}$
- *Conditional probability (given the first was red):* $P(B|A) = \frac{3}{5}$

Solution: Non-Independence (4/4)

Solution

Comparing $P(B)$ and $P(B|A)$:

Let's look at the two numbers we just calculated:

- *Unconditional probability of drawing a red sock second:* $P(B) = \frac{2}{3}$
- *Conditional probability (given the first was red):* $P(B|A) = \frac{3}{5}$

The Core Definition of Dependence

Because $P(B) \neq P(B|A)$, the outcome of the first draw physically changes the probability of the second. Therefore, events A and B are ***dependent****.*

Summary: Probability Fundamentals

- **The Language of Probability:** All experiments have a sample space (S). Events are subsets of that space.
- **The Addition Law (Unions):** To find the probability of $A \cup B$, add their probabilities but remember to subtract the intersection!

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- **The Multiplication Law (Intersections):** To find the probability of $A \cap B$, multiply $P(A)$ by the conditional probability of B given A.

$$P(A \cap B) = P(A) \times P(B|A)$$

- **Independence:** If A happening doesn't change the probability of B happening, they are independent $\Rightarrow P(B|A) = P(B)$.

Looking Ahead

Next Up: Discrete Probability Distributions

Now that we know the basic rules of random events, we will map these events to numerical variables. Instead of just looking at 'Pass/Fail' events, we will learn how to calculate the probability of getting *exactly 3 failures* in a batch of 50!

Block III

Discrete Probability Distributions

13 Discrete Probability Distributions

Outline - Next Section I

13 Discrete Probability Distributions
- The Binomial Distribution
- The Poisson Distribution

From Events to Random Variables

In the previous lecture, we looked at the probability of specific, single events (like drawing a red sock). In engineering, we usually want to model a whole system of events mathematically.
To do this, we use **Random Variables**.

Definition (Discrete Random Variable)

A discrete random variable (usually denoted by a capital letter like X) is a function that assigns a **numerical value** to each outcome in a sample space. It can only take on a countable number of distinct values (e.g., $0, 1, 2, 3 \dots$).

Discrete Random Variables: Expectation I

Before we look at complex models, we must understand how to find the **average** outcome of a random process.

Definition (Expected Value $E(X)$)

The Expected Value (or mean) of a discrete random variable is the weighted average of all possible outcomes:

$$E(X) = \mu = \sum_{i=1}^{n} x_i P(x_i)$$

Discrete Random Variables: Expectation II

Example (Component Testing)

A technician tests a batch of 3 sensors. The probability of x sensors being faulty is given below:

x	0	1	2	3
$P(X = x)$	0.70	0.20	0.08	0.02

Calculate the expected number of faulty sensors, $E(X)$.

Solution: Expectation

Solution

To find $E(X)$, we multiply each outcome by its probability and sum them up:

$$
\begin{aligned}
E(X) &= (0 \times 0.70) + (1 \times 0.20) + (2 \times 0.08) + (3 \times 0.02) \\
&= 0 + 0.20 + 0.16 + 0.06 \\
&= \mathbf{0.42} \textit{ sensors}
\end{aligned}
$$

Engineering Interpretation: *If we tested thousands of batches, we would find an average of 0.42 faults per batch.*

The Binomial Distribution I

The Binomial Distribution is the ultimate tool for Quality Assurance engineers. It models the number of **successes** in a fixed number of **independent trials**.

Conditions for a Binomial Model

To use this model, your experiment must meet four strict rules:

1. There is a fixed number of trials (n).
2. Each trial has only two outcomes ('Success' or 'Failure').
3. The probability of success (p) is the same for every trial.
4. The trials are completely **independent**.

The Binomial Distribution II

Definition (The Binomial Formula)

If $X \sim \text{Bin}(n, p)$, the probability of getting exactly k successes is:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Where $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ *is the number of possible combinations.*

The '8 Heads' Problem Solved I

Example (Is the coin biased?)

In Part 1, we asked: *'You flip a fair coin 10 times. You should get 5 heads. You actually get 8. Is the coin biased, or is it just bad luck?'* Now we have the exact formula to solve this. Assume the coin is perfectly fair ($p = 0.5$). Let X be the number of heads in $n = 10$ independent flips.

Question:

1. Calculate the exact probability of getting **exactly 8** heads by pure random chance.
2. **If** each person in a class of 300 students was to do this experiment, how many would get exactly 8 heads?

The '8 Heads' Problem Solved II

Solution

1. Here, $X \sim Bin(10, 0.5)$*. We want to find* $P(X = 8)$*.*

Calculation:

$$\begin{aligned} P(X = 8) &= \binom{10}{8} (0.5)^8 (1 - 0.5)^{10-8} \\ &= \frac{10!}{8!(2)!} (0.5)^8 (0.5)^2 \\ &= 45 \times (0.5)^{10} \\ &= 45 \times 0.000976 \\ &= \mathbf{0.0439} \textit{ (or 4.39\%)} \end{aligned}$$

2. ***Expected number of students:***

$$E(X) = 300 \times 0.0439 \approx \mathbf{13} \textit{ students}$$

Engineering Example: Quality Control I

Example (Manufacturing Yield)

A factory produces LED bulbs. Historically, 5% of the bulbs are defective ($p = 0.05$). An engineer pulls a random sample of 10 bulbs off the assembly line ($n = 10$).

Questions:

1. What is the probability that **exactly 0** bulbs are defective? (A perfect batch).
2. What is the probability that **exactly 2** bulbs are defective?

Engineering Example: Quality Control II

Solution

Let X be the number of defective bulbs. $X \sim Bin(10, 0.05)$.

1. Exactly 0 defective $(k = 0)$***:***

$$P(X = 0) = \binom{10}{0}(0.05)^0(1 - 0.05)^{10-0}$$
$$= 1 \times 1 \times (0.95)^{10}$$
$$= \mathbf{0.5987} \textit{ (or 59.87\%)}$$

Engineering Example: Quality Control III

Solution

2. Exactly 2 defective ($k = 2$):

$$
\begin{aligned}
P(X = 2) &= \binom{10}{2}(0.05)^2(0.95)^8 \\
&= \frac{10!}{2!(8)!} \times 0.0025 \times 0.6634 \\
&= 45 \times 0.0025 \times 0.6634 \\
&= \mathbf{0.0746} \textit{ (or 7.46\%)}
\end{aligned}
$$

Mean and Variance of a Binomial Model

Because the Binomial distribution is highly structured, calculating its expected value and spread doesn't require building complex tables!

Binomial Shortcuts

If $X \sim \text{Bin}(n, p)$:

- **Mean (Expectation):** $\mu = E(X) = np$
- **Variance:** $\sigma^2 = np(1-p)$
- **Standard Deviation:** $\sigma = \sqrt{np(1-p)}$

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- **Variance:** $\sigma^2 = np(1-p)$
- **Standard Deviation:** $\sigma = \sqrt{np(1-p)}$

Example (Expected Failures)

If we sample 10 bulbs with a 5% defect rate, how many do we *expect* to be defective?

$$E(X) = 10 \times 0.05 = \mathbf{0.5} \text{ bulbs}$$

Python: The Binomial Distribution

Calculating factorials by hand is fine for $n = 10$, but not practical for $n = 10,000$. Engineers use the `scipy.stats` library to calculate complex probabilities instantly.

Solving the '8 Heads' Problem: We want exactly 8 heads from 10 flips ($p = 0.5$). We use the Probability Mass Function (`pmf`).

Our 8 heads python style

```python
from scipy.stats import binom

# binom.pmf(k, n, p)
p_8_heads = binom.pmf(8, 10, 0.5)

print(f"Probability of 8 heads: {p_8_heads:.4f}")
# Output: 0.0439
```

Notice that Python gives us the exact same 4.39% we calculated by hand!

The Poisson Distribution

While the Binomial distribution counts successes in a *fixed number of trials*, the **Poisson Distribution** counts events that occur randomly over a **continuous interval** (like time, area, or volume).

When to use Poisson in EEE:

- Number of network packets lost per minute.
- Number of microscopic defects per square meter of silicon.
- Number of server crashes per month.

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When to use Poisson in EEE:

- Number of network packets lost per minute.
- Number of microscopic defects per square meter of silicon.
- Number of server crashes per month.

Definition (The Poisson Formula)

If $X \sim \text{Pois}(\lambda)$, the probability of exactly k events occurring is:

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$$

Where λ (lambda) is the average rate of occurrence, and $e \approx 2.718$.

Engineering Example: Network Faults I

Example (Fiber Optic Cable Faults)

A long-haul fiber optic network experiences an average of $\lambda = 2$ dropped connections per day.

Questions:

1. What is the probability that the network experiences **exactly 0** drops tomorrow? (A perfect day).
2. What is the probability that the network experiences **at most 2** drops?

Engineering Example: Network Faults II

Solution

Let X be the number of dropped connections per day. $X \sim Pois(2)$.

1. Exactly 0 drops ($k = 0$):

$$\begin{aligned} P(X = 0) &= \frac{e^{-2}2^0}{0!} \\ &= \frac{0.1353 \times 1}{1} \\ &= \mathbf{0.1353} \text{ \textit{(or 13.5\% chance of a perfect day)}} \end{aligned}$$

Engineering Example: Network Faults III

Solution

2. At most 2 drops ($X \leq 2$):

This means 0 drops, 1 drop, ***or*** *2 drops. We must add the probabilities!*

$$\begin{aligned} P(X \leq 2) &= P(X = 0) + P(X = 1) + P(X = 2) \\ &= e^{-2}\left(\frac{2^0}{0!} + \frac{2^1}{1!} + \frac{2^2}{2!}\right) \\ &= 0.1353 \times \left(1 + 2 + \frac{4}{2}\right) \\ &= 0.1353 \times 5 \\ &= \mathbf{0.6765} \textit{ (or 67.65\%)} \end{aligned}$$

Mean and Variance of a Poisson Model

The Poisson distribution has a unique and beautiful mathematical property regarding its average and its spread.

Poisson Shortcuts

If $X \sim \text{Pois}(\lambda)$:

- **Mean (Expectation):** $\mu = E(X) = \lambda$
- **Variance:** $\sigma^2 = \lambda$

Mean and Variance of a Poisson Model

The Poisson distribution has a unique and beautiful mathematical property regarding its average and its spread.

Poisson Shortcuts

If $X \sim \text{Pois}(\lambda)$:

- **Mean (Expectation):** $\mu = E(X) = \lambda$
- **Variance:** $\sigma^2 = \lambda$

The Mathematical Oddity

In a Poisson distribution, the mean is **exactly equal** to the variance! As the average rate of events increases, the uncertainty (spread) around that rate increases at the exact same pace.

Python: The Poisson Distribution

Solving the 'At most 2 drops' Problem: Instead of manually adding $P(X=0)+P(X=1)+P(X=2)$, we can use the Cumulative Distribution Function (`cdf`) to add them automatically!

Poisson Example

```
from scipy.stats import poisson

# 1. Exactly 0 drops (PMF)
p_0 = poisson.pmf(0, 2)
print(f"Exactly 0 drops: {p_0:.4f}")
# Output: 0.1353

# 2. At most 2 drops (CDF)
p_up_to_2 = poisson.cdf(2, 2)
print(f"At most 2 drops: {p_up_to_2:.4f}")
# Output: 0.6767
```

Summary: Discrete Distributions

- **Random Variables (X)**
  - Maps sample spaces to numerical values.
  - **Expectation:** $E(X) = \sum xP(x)$ (The long-run average).

- **The Binomial Model**
  - Conditions: Fixed n, Binary outcomes (p/q), Independence.
  - Formula: $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$
  - Mean $\mu = np$, Variance $\sigma^2 = np(1-p)$.

- **The Poisson Distribution**
  - Modelling rare events over a continuous interval (time/space).
  - Formula: $P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$
  - Mean and Variance are identical: $\mu = \sigma^2 = \lambda$.

Summary: Discrete Distributions

- **Random Variables (X)**
  - Maps sample spaces to numerical values.
  - **Expectation:** $E(X) = \sum xP(x)$ (The long-run average).

- **The Binomial Model**
  - Conditions: Fixed n, Binary outcomes (p/q), Independence.
  - Formula: $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$
  - Mean $\mu = np$, Variance $\sigma^2 = np(1-p)$.

- **The Poisson Distribution**
  - Modelling rare events over a continuous interval (time/space).
  - Formula: $P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$
  - Mean and Variance are identical: $\mu = \sigma^2 = \lambda$.

Next Up: Continuous Probability Distributions

But what happens when our random variable is a continuous physical measurement?

MATH19622-1E2: Introduction to Probability and Statistics

4 Lecture Series (90 minutes per Lecture)

Continuous Distributions and The Normal Curve

Anthony R. Thornton

2025/2026

Block IV

Continuous Probability Distributions

Outline - Next Section I

From Counting to Measuring

So far, we have looked at **Discrete** variables: things we can count (e.g., $0, 1, 2$ defective chips).

But what if we are measuring the exact voltage of a 5V power supply? It might be 5.01V, or 5.013V, or 5.01347V ...

Definition (Continuous Random Variable)

A continuous random variable can take on *any* real value within a given range. Because there are infinitely many possible values, we cannot list them in a table or use a Probability Mass Function.

The Probability Density Function (PDF)

Continuous variables are defined a **Probability Density Function**, or PDF, usually denoted as $f(x)$.

Two Rules of a PDF

1. The curve must never dip below the x-axis: $f(x) \geq 0$.
2. The **total area** under the entire curve must equal exactly 1 (or 100%).

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1. The curve must never dip below the x-axis: $f(x) \geq 0$.
2. The **total area** under the entire curve must equal exactly 1 (or 100%).

The Golden Rule of Continuous Probability

To find the probability that a value falls between a and b, you calculate the area under the curve between those two points:

$$P(a \leq X \leq b) = \int_a^b f(x)\,dx$$

The ‘Exact Point’ Paradox

Example (The infinitely precise resistor)

Suppose you have a resistor that is supposedly exactly $100\,\Omega$. What is the probability that its true, physical resistance is *exactly* $100.000000000\ldots\,\Omega$ with infinite precision?

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Solution

Mathematically, a single point has a width of zero. Therefore, the area under the curve at exactly one point is zero!

$$P(X = 100) = \int_{100}^{100} f(x)\,dx = \mathbf{0}$$

Engineering Takeaway: *For continuous variables, it is meaningless to ask for the probability of an exact number. We can only ask for the probability that a value falls within a* ***tolerance range*** *(e.g., between* $99.9\,\Omega$ *and* $100.1\,\Omega$*).*

The Cumulative Distribution Function (CDF)

Definition (The CDF)

The CDF gives the accumulated area under the PDF curve from negative infinity up to a specific point x:

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)\,dt$$

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Why the CDF is powerful

If you already know the CDF, finding the probability of a value falling within a range becomes simple subtraction, with no integration required!

$$P(a \leq X \leq b) = F(b) - F(a)$$

Mean and Variance of Continuous Distributions

Expected Value (Mean)

The expected value, μ or $E(X)$, represents the 'centre of mass' of the distribution:

$$\mu = E(X) = \int_{-\infty}^{\infty} x f(x) \, \mathrm{d}x$$

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Variance

The variance, σ^2 or $V(X)$, measures the spread around the mean:

$$\sigma^2 = V(X) = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\,\mathrm{d}x$$

$$V(X) = E(X^2) - [E(X)]^2 \quad \text{where} \quad E(X^2) = \int_{-\infty}^{\infty} x^2 f(x)\,\mathrm{d}x$$

Worked Example: A Custom Continuous Distribution

Example (Sensor Calibration Error)

An engineer models the calibration error X (in millivolts) of a new prototype thermal sensor using the following Probability Density Function (PDF):

$$f(x) = \begin{cases} cx & \text{for } 0 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

Questions:

1. Find the constant c that makes this a valid PDF.
2. Determine the Cumulative Distribution Function (CDF), $F(x)$.
3. What is the probability that the error is less than 1 mV?
4. Calculate the expected (mean) error, $E(X)$.
5. Calculate the variance (spread).

Solution: A Custom Continuous Distribution (1/4)

Solution

1. Find the constant c***:***

The total area under the PDF must equal exactly 1.

$$\int_0^2 cx \, \mathrm{d}x = 1$$

$$\left[c\frac{x^2}{2}\right]_0^2 = 1$$

$$c\left(\frac{4}{2} - 0\right) = 1 \implies 2c = 1 \implies \mathbf{c = 0.5}$$

So, our complete PDF is $f(x) = 0.5x$ *(for* $0 \leq x \leq 2$*).*

Solution: A Custom Continuous Distribution (2/4)

Solution

2. Determine the CDF, $F(x)$***:***

The CDF accumulates area from the lower bound (0) *up to an arbitrary point* x. *We integrate using a dummy variable* t*:*

$$\begin{aligned} F(x) &= \int_0^x 0.5t \, \mathrm{d}t \\ &= \left[0.25t^2\right]_0^x = \mathbf{0.25x^2} \end{aligned}$$

(Valid for $0 \leq x \leq 2$. *Once* x *passes 2, the CDF stays flat at 1).*

Solution: A Custom Continuous Distribution (3/4)

Solution

3. Probability the error is less than* 1 *mV* ($P(X < 1)$)*:

Because we already found the CDF, we don't need to integrate again! We simply plug 1 *into* $F(x)$*:*

$$P(X < 1) = F(1) = 0.25(1)^2 = \mathbf{0.25} \textbf{\textit{ (or 25\%)}}$$

Solution: A Custom Continuous Distribution (4/4)

Solution

4. Expected Value (Mean), $E(X)$***:***

For continuous variables, the sum becomes an integral:
$E(X) = \int x f(x)\,\mathrm{d}x$.

$$\begin{aligned} E(X) &= \int_0^2 x(0.5x)\,\mathrm{d}x \\ &= \int_0^2 0.5x^2\,\mathrm{d}x \\ &= \left[\frac{0.5x^3}{3}\right]_0^2 = \frac{4}{3} \approx \mathbf{1.33}\ \boldsymbol{mV} \end{aligned}$$

Solution: A Custom Continuous Distribution (5/5)

Solution

5. Calculate the Variance, $V(X)$***:***

$$
\begin{aligned}
E(X^2) &= \int_0^2 x^2 f(x)\,\mathrm{d}x \\
&= \int_0^2 x^2(0.5x)\,\mathrm{d}x = \int_0^2 0.5x^3\,\mathrm{d}x \\
&= \left[\frac{0.5x^4}{4}\right]_0^2 = \frac{0.5(16)}{4} = \mathbf{2}
\end{aligned}
$$

$$
\begin{aligned}
V(X) &= 2 - \left(\frac{4}{3}\right)^2 \\
&= 2 - \frac{16}{9} = \frac{18}{9} - \frac{16}{9} = \boldsymbol{\frac{2}{9}} \approx \boldsymbol{0.222\ mV^2}
\end{aligned}
$$

Outline - Next Section I

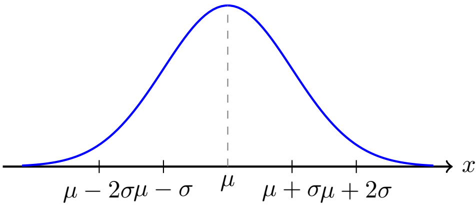
The Normal (Gaussian) Distribution

It naturally models thermal noise in circuits, manufacturing tolerances, and measurement errors.

If a variable X follows a normal distribution, we write: $X \sim N(\mu, \sigma^2)$
Where μ is the mean (the centre) and σ^2 is the variance (the spread).

Properties

- **Bell-Shaped & Symmetric**
- **Centered:** Mean = Median = Mode = μ.
- **Asymptotic:** Tails never touch the x-axis.



The Equation of the Normal Curve

The bell shape we just saw is not just a rough sketch; it is defined by a very specific, and slightly intimidating, mathematical formula.

Definition (Normal Probability Density Function)

If a continuous random variable X is normally distributed with mean μ and standard deviation σ, its PDF is given by:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

for any real number x $(-\infty < x < \infty)$.

Example

Question: What is the CDF for the normal curve? How do we integrate the PDF to get the CDF for the normal curve?

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Integrating the normal distribution

Solution

This integral

$$\int_{-\infty}^{x} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \mathrm{d}x$$

can only be done numerically

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can only be done numerically

Why do we use Python (or Tables)?

Remember that probability is the **area** under this curve. However, the integral of this specific function **cannot be solved algebraically**! There is no closed-form solution.
We must use numerical approximation (like Python's `scipy.stats.norm.cdf`) to find the areas.

The CDF and the Error Function (erf)

As we saw, the integral for the Normal CDF cannot be solved using standard algebra. To evaluate it, mathematicians defined a special, non-elementary function called the **Error Function** (erf).

Definition (The Error Function)

$$\operatorname{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-t^2}\, dt$$

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Definition (The Error Function)

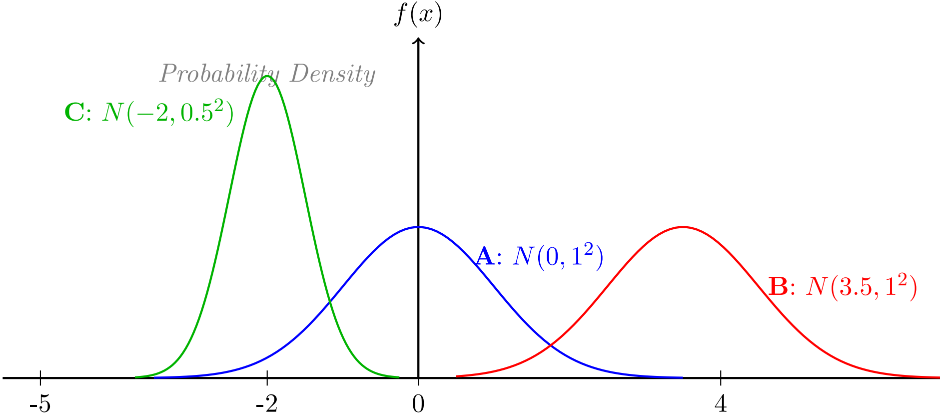
$$\mathrm{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-t^2}\, dt$$

The exact Cumulative Distribution Function (CDF) of any Normal Distribution can be expressed using this function:

$$F(x) = P(X \leq x) = \frac{1}{2}\left[1 + \mathrm{erf}\left(\frac{x-\mu}{\sigma\sqrt{2}}\right)\right]$$

Outline - Next Section I

The Infinite Family of Normal Curves



A is centered at 0. **B** has the same shape, but is shifted.
C is tall and skinny because its standard deviation is small.

Standardising the Normal Curve

Before computers existed, engineers couldn't calculate the area under every possible curve. Instead, they used a mathematical trick called **Standardisation** to convert *any* normal distribution into one single, universal curve.

Definition (The Z-Score)

A Z-score transforms a raw measurement (X) into a standardised number by subtracting the mean and dividing by the standard deviation:

$$Z = \frac{X - \mu}{\sigma}$$

The Standard Normal Distribution

When you apply the Z-score formula to every single data point in a dataset, you create a brand new distribution called the **Standard Normal Distribution**.

Properties of the Standard Normal (Z)

- The mean is always shifted to zero: $\mu_Z = 0$
- The standard deviation is always scaled to one: $\sigma_Z = 1$
- Notation: $Z \sim N(0, 1)$

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The Golden Rule of Z-Scores

The area under the curve for the original X value is **exactly the same** as the area under the curve for the new Z value.

$$P(X \leq x) = P\left(Z \leq \frac{x - \mu}{\sigma}\right)$$

Example: Comparing Apples to Oranges I

Z-scores aren't just an old trick for using lookup tables; they are the ultimate engineering tool for comparing different systems!

Example: Comparing Apples to Oranges II

Example (Quality Control on Two Lines)

You are managing two different manufacturing lines.

- **Line A (Resistors):** $\mu = 1000\,\Omega$, $\sigma = 20\,\Omega$.
- **Line B (Capacitors):** $\mu = 50\,\mu\text{F}$, $\sigma = 0.5\,\mu\text{F}$.

An inspector pulls one component from each line:

- The resistor measures $1035\,\Omega$.
- The capacitor measures $51.2\,\mu\text{F}$.

Question: Which of these two components is more "unusual" (i.e., relatively further from its design specification)?

Example: Comparing Apples to Oranges III

Solution

We cannot compare Ohms to Microfarads directly. We must convert both to unitless Z-scores!

1. Line A (Resistor):

$$Z_A = \frac{X - \mu}{\sigma} = \frac{1035 - 1000}{20} = \frac{35}{20} = \mathbf{1.75}$$

The resistor is 1.75 *standard deviations above its mean.*

2. Line B (Capacitor):

$$Z_B = \frac{X - \mu}{\sigma} = \frac{51.2 - 50}{0.5} = \frac{1.2}{0.5} = \mathbf{2.40}$$

The capacitor is 2.40 *standard deviations above its mean.*

The Standard Normal Table (Z-Table)

x	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9773	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.6	0.9998	0.9998	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.7	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.8	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

How to Read a Z-Table

Reading the Table

- **Left Column:** Gives the Z-score up to the first decimal place (e.g., 1.2).
- **Top Row:** Gives the second decimal place (e.g., 0.04).
- **Intersection:** The value at row 1.2 and column 0.04 gives the total area: $P(Z \leq 1.24)$.

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- **Intersection:** The value at row 1.2 and column 0.04 gives the total area: $P(Z \leq 1.24)$.

Handling Negative Z-Scores

Notice your table only shows positive Z values! Because the Normal Curve is perfectly symmetric, the left tail is identical to the right tail. To find negative areas, we use the complement rule:

$$P(Z \leq -z) = 1 - P(Z \leq z)$$

Worked Example: Calculating Probabilities

Example (Resistor Manufacturing)

A production line manufactures resistors with a mean resistance of $\mu = 1000\,\Omega$ and a standard deviation of $\sigma = 20\,\Omega$.

Assume the resistance values are normally distributed. Using your Z-table handout, calculate:

1. The probability that a randomly selected resistor is **less than** $1035\,\Omega$.
2. The probability that a resistor is **greater than** $1025\,\Omega$.
3. The probability that a resistor is **less than** $970\,\Omega$.

Solution: Using the Z-Table (1/2)

Solution

1. Less than $1035\,\Omega$***:***

- *Find the Z-score:* $Z = \frac{1035-1000}{20} = 1.75$
- *We want* $P(Z \leq 1.75)$*. This is a direct lookup.*
- *Look at row* ***1.7****, column* ***0.05****.*
- *Table value:* **0.9599** *(or 95.99%).*

Solution: Using the Z-Table (1/2)

Solution

1. Less than $1035\,\Omega$***:***

- *Find the Z-score:* $Z = \frac{1035-1000}{20} = 1.75$
- *We want* $P(Z \leq 1.75)$*. This is a direct lookup.*
- *Look at row **1.7**, column **0.05**.*
- *Table value:* $\mathbf{0.9599}$ *(or 95.99%).*

2. Greater than $1025\,\Omega$***:***

- *Find the Z-score:* $Z = \frac{1025-1000}{20} = 1.25$
- *We want* $P(Z > 1.25)$*. The table only gives "less than" areas!*
- *Use the complement:* $P(Z > 1.25) = 1 - P(Z \leq 1.25)$*.*
- *Look at row **1.2**, column **0.05**. Table value is* 0.8944*.*
- *Answer:* $1 - 0.8944 = \mathbf{0.1056}$ *(or 10.56%).*

Solution: Using the Z-Table (2/2)

Solution

3. Less than $970\,\Omega$***:***

- *Find the Z-score:* $Z = \frac{970-1000}{20} = \frac{-30}{20} = -1.50$
- *We want* $P(Z \leq -1.50)$*. Our table has no negative numbers!*
- *By symmetry, the extreme left tail area equals the extreme right tail area:*
  $P(Z \leq -1.50) = P(Z \geq 1.50) = 1 - P(Z \leq 1.50)$
- *Look at row* ***1.5****, column* ***0.00****. Table value is* 0.9332*.*
- *Answer:* $1 - 0.9332 = \mathbf{0.0668}$ *(or 6.68%).*

The Empirical Rule (68-95-99.7)

For *any* Normal Distribution:

- $\approx 68\%$ of the data falls within **1 standard deviation** of the mean ($\mu \pm 1\sigma$).
- $\approx 95\%$ of the data falls within **2 standard deviations** of the mean ($\mu \pm 2\sigma$).
- $\approx 99.7\%$ of the data falls within **3 standard deviations** of the mean ($\mu \pm 3\sigma$).

Example: The Empirical Rule

Example (Manufacturing Tolerances)

A factory produces 9V batteries. The actual voltage is normally distributed with a mean of $\mu = 9.0$V and a standard deviation of $\sigma = 0.1$V.

Questions:

1. What percentage of batteries will have a voltage between 8.9V and 9.1V?
2. What percentage of batteries will have a voltage between 8.8V and 9.2V?
3. If the battery is considered 'defective' if it falls outside of the 8.7V to 9.3V range, what is the defect rate?

Solution: The Empirical Rule

Solution

1. *8.9 V to 9.1 V is exactly $\mu - 1\sigma$ to $\mu + 1\sigma$.*
   According to the rule, this covers $\approx 68\%$ of the batteries.

2. *8.8 V to 9.2 V is exactly $\mu - 2\sigma$ to $\mu + 2\sigma$.*
   This covers $\approx 95\%$ of the batteries.

3. *8.7 V to 9.3 V represents the $\pm 3\sigma$ range.*
   We know $\approx 99.7\%$ of batteries fall inside this safe range.
   Therefore, the defect rate is everything outside this range:
   $100\% - 99.7\% = \mathbf{0.3}\%$ ***defect rate***.

Python: The Normal Distribution

Solving the Battery Problem (8.8V to 9.2V):

Normal Distribution in Python

```python
from scipy.stats import norm
# Define our battery parameters
mu = 9.0        # Mean (loc)
sigma = 0.1     # Standard Deviation (scale)
# Find the area up to the upper limit (9.2V)
upper_area = norm.cdf(9.2, loc=mu, scale=sigma)
# Find the area up to the lower limit (8.8V)
lower_area = norm.cdf(8.8, loc=mu, scale=sigma)
# The probability of being between 8.8V and 9.2V
prob_between = upper_area - lower_area
print(f"Probability between 8.8V and 9.2V: {prob_between:.4f}")
# Output: Probability between 8.8V and 9.2V: 0.9545 (or 95.45%)
```

The Grand Convergence: Large Numbers

What happens to our discrete distributions when we deal with massive quantities (like millions of transistors or network packets)?

Binomial to Normal

When n is very large:
$X \sim \text{Bin}(n, p) \approx N(np, np(1-p))$

Parameters:

- $\mu = np$
- $\sigma^2 = np(1-p)$

Rule of thumb: $np > 5$ *and* $n(1-p) > 5$

Poisson to Normal

When the rate λ is very large:
$X \sim \text{Pois}(\lambda) \approx N(\lambda, \lambda)$

Parameters:

- $\mu = \lambda$
- $\sigma^2 = \lambda$

Rule of thumb: $\lambda > 20$

Example: Binomial to Normal I

Example (Manufacturing Yield)

A factory produces memory chips with a historical defect rate of 20% ($p = 0.2$). Today, they manufactured 400 chips.
Using the Normal approximation, what is the probability that they produced **less than 64** defective chips?
Comment on the how your would do the exact calculation?

Example: Binomial to Normal II

Solution

Step 1: Check the rules and find the parameters.

- *Check:* $np = 400 \times 0.2 = 80$ *(Greater than 5! ✓)*
- *Mean:* $\mu = np = 80$ *defects.*
- *Variance:* $\sigma^2 = np(1-p) = 80 \times 0.8 = 64$.
- *Standard Deviation:* $\sigma = \sqrt{64} = \mathbf{8}$ *defects.*

Step 2: Convert to a Z-score. *We want* $P(X < 64)$.

$$Z = \frac{X-\mu}{\sigma} = \frac{64-80}{8} = \frac{-16}{8} = \mathbf{-2.00}$$

Step 3: Look up the area. *Using our symmetric Z-table:*
$P(Z < -2.00) = 1 - P(Z \leq 2.00)$.
Table lookup for 2.00 *is* 0.9772.
Answer: $1 - 0.9772 = \mathbf{0.0228}$ ***(or 2.28%)***.

Solution

Step 4: The Reality Check (Exact vs. Approx)

If we asked a computer to calculate the exact, discrete Binomial probability, it would have to sum 64 massive factorial calculations:

$$P(X < 64) = P(X \leq 63) = \sum_{x=0}^{63} \binom{400}{x} (0.2)^x (0.8)^{400-x} = \mathbf{0.0195} \ \boldsymbol{(1.95\%}$$

Comparison

Our quick Normal approximation (2.28%) is very close to the exact value (1.95%), but it is off by about 0.33%.

Note: If we apply a ± 0.5 Continuity Correction and look up $P(X < 63.5)$ instead, our Z-score becomes -2.06, giving an approximation of ***1.97%****—an almost perfect match!*

Example: Poisson to Normal I

Example (Fiber Optic Packets)

A fibre optic router receives an average of 100 data packets per millisecond ($\lambda = 100$).
Using the Normal approximation, what is the probability that the router receives **more than 120** packets in the next millisecond?
Comment of how you would do the exact calculation?

Example: Poisson to Normal II

Solution

Step 1: Check the rules and find the parameters.

- *Check:* $\lambda = 100$ *(Greater than 20! ✓)*
- *Mean:* $\mu = \lambda = 100$ *packets.*
- *Variance:* $\sigma^2 = \lambda = 100$.
- *Standard Deviation:* $\sigma = \sqrt{100} = \mathbf{10}$ *packets.*

Step 2: Convert to a Z-score. *We want* $P(X > 120)$.

$$Z = \frac{X - \mu}{\sigma} = \frac{120 - 100}{10} = \frac{20}{10} = \mathbf{2.00}$$

Solution

Step 3: Look up the area. *Because we want 'more than', we use the complement rule:*
$P(Z > 2.00) = 1 - P(Z \leq 2.00)$.
Table lookup for 2.00 *is* 0.9772.
Answer: $1 - 0.9772 =$ **0.0228 *(or 2.28%)***.
Step 4: The Reality Check (Exact vs. Approx)

$$P(X > 120) = 1 - \sum_{x=0}^{120} \frac{e^{-100} 100^x}{x!} = \mathbf{0.0227} \ \textbf{\textit{(2.27\%)}}$$

Comparison

Our hand-calculated Normal approximation (2.28%) is accurate to the exact Poisson value (2.27%) to within one-hundredth of a percent! The approximation performs incredibly well when λ *is large.*

Beyond the Normal Curve

The Normal distribution is everywhere, but it is not the only continuous probability model! Depending on your engineering discipline, you will eventually encounter others:

- **Uniform Distribution:** A flat rectangle where any value in a range is equally likely.
  - *Used in:* Modeling quantization error in Analog-to-Digital Converters (ADCs).
- **Exponential Distribution:** Models the time between independent events.
  - *Used in:* Reliability engineering (Mean Time Between Failures) and network queueing delays.
- **Weibull & Rayleigh Distributions:** Skewed distributions that model asymmetric continuous data.
  - *Used in:* Predicting material fatigue life in mechanical engineering, or modeling wireless signal fading in communications.

Formulas for Other Distributions I

While not strictly on this semester's syllabus, here is the mathematical reference for the distributions we just mentioned. Notice how every single one still has a clearly defined PDF, mean, and variance!

Continuous Uniform Distribution: $U(a, b)$

Models an equal probability across a specific range from a to b.

- **PDF:** $f(x) = \frac{1}{b-a} \quad \text{for } a \leq x \leq b$
- **Mean:** $\mu = \frac{a+b}{2}$
- **Variance:** $\sigma^2 = \frac{(b-a)^2}{12}$

Formulas for Other Distributions II

Exponential Distribution: $\mathrm{Exp}(\lambda)$

Models the time between events, using a rate parameter λ.

- **PDF:** $f(x) = \lambda e^{-\lambda x} \quad \text{for } x \geq 0$
- **Mean:** $\mu = \frac{1}{\lambda}$
- **Variance:** $\sigma^2 = \frac{1}{\lambda^2}$

Formulas for Other Distributions III

Rayleigh Distribution

Models the overall magnitude of a vector when its components are independent and normally distributed (like 2D wind velocity or radio signal scattering).

- **PDF:** $f(x) = \frac{x}{\sigma^2} e^{-x^2/(2\sigma^2)} \quad \text{for } x \geq 0$
- **Mean:** $\mu = \sigma\sqrt{\frac{\pi}{2}}$
- **Variance:** $V(X) = \frac{4-\pi}{2}\sigma^2$

Formulas for Other Distributions IV

Weibull Distribution (2-Parameter)

A highly versatile distribution heavily used in reliability engineering, featuring a shape parameter k and a scale parameter λ.

- **PDF:** $f(x) = \frac{k}{\lambda}\left(\frac{x}{\lambda}\right)^{k-1} e^{-(x/\lambda)^k} \quad \text{for } x \geq 0$
- **Mean:** $\mu = \lambda\Gamma\left(1 + \frac{1}{k}\right)$
- *Note:* Γ *represents the Gamma function. The exact variance involves multiple Gamma terms!*